

HR Analytics Dashboard Project

Employee Attrition Analysis & Retention Strategy

Project Overview

The HR Analytics Dashboard is a comprehensive Power BI solution designed to analyze employee attrition patterns and support data-driven retention strategies. This project leverages advanced analytics to identify key factors contributing to the 16.28% organization-wide attrition rate and provides actionable insights for executive and HR teams to implement targeted interventions.

Project Objectives

- Analyze attrition patterns across departments, roles, and demographics
- Identify correlation between work factors (overtime, satisfaction) and employee retention
- Create interactive visualizations for real-time attrition monitoring
- Develop predictive models to identify flight-risk employees
- Generate actionable recommendations for HR strategy

Dataset Specifications

Source: HR employee database
Records: 1,480 employees
Attributes: 37 features across demographics, job characteristics, and satisfaction metrics
Time Period: Historical snapshot with attrition outcomes
Data Quality: Cleaned and validated for analysis

Key Dataset Metrics

Metric	Value
Total Employees	1,480
Employees Retained	1,239 (83.72%)
Employees Lost	241 (16.28%)
Average Monthly Income	\$49,511
Salary Range	\$1,055 - \$99,806
Average Employee Age	36.9 years
Departments Covered	7
Job Roles Analyzed	9+

Dashboard Components

1. Executive Summary Dashboard

- Overall attrition rate and trends
- Key performance indicators (KPIs)
- Department-level attrition comparison

2. Departmental Analysis

- Attrition by department with rankings
- Headcount and retention metrics
- Department-specific trends

3. Employee Segmentation

- Demographics analysis (age, gender, marital status)
- Tenure and experience patterns
- Salary bracket analysis

4. Attrition Drivers

- Overtime impact on retention
- Job satisfaction correlations
- Work-life balance metrics
- Job involvement and environment satisfaction

5. Role-Based Analysis

- Attrition by job role
- Career progression patterns
- Role-specific retention strategies

Technologies & Tools

Component	Technology
Dashboard Platform	Microsoft Power BI
Data Source	CSV file (HR Database)
Programming	Python (Analysis & Preprocessing)
Analysis	Pandas, NumPy, Scikit-learn
Visualization	Power BI Custom Visuals
Version Control	GitHub

Key Findings

Critical Insight - Overtime Impact:

Employees working overtime show a 31.19% attrition rate, nearly 3x higher than non-overtime workers (10.38%), indicating overtime as a critical retention risk factor.

Departmental Variations:

- Highest Risk: Human Resources (26.83%), Sales (25.23%)
- Lowest Risk: Operations (8.33%)
- Above Average: Administration (20.71%)

Compensation Paradox:

Interestingly, employees earning higher salaries show similar or slightly higher attrition, suggesting

non-monetary factors drive attrition decisions.

Demographic Patterns:

Gender distribution (Male: 888, Female: 592) shows relatively similar attrition patterns, indicating attrition is not primarily gender-driven.

Recommendations

1. Overtime Management Strategy

- Implement mandatory overtime limits
- Provide overtime compensation and flexibility options
- Monitor overtime impact on individual retention

2. Department-Specific Interventions

- Develop targeted retention programs for HR and Sales
- Learn from Operations' best practices
- Implement role-specific career development

3. Work Environment Improvement

- Enhance job satisfaction initiatives
- Improve work-life balance policies
- Increase environment satisfaction through workplace improvements

4. Retention Monitoring

- Use dashboard for proactive flight-risk identification
- Implement early intervention programs
- Track retention program effectiveness

5. Career Development

- Create clear career progression paths
- Implement mentorship programs
- Provide skills development opportunities

Project Deliverables

- ✓ HR Analytics Dashboard (Power BI)
- ✓ Dataset with preprocessing scripts (Python)
- ✓ Executive summary reports
- ✓ Documentation and README
- ✓ GitHub repository with full project structure

Project Created: June 2026

For detailed analysis and interactive visualizations, refer to the Power BI dashboard.